AI Agent Studio. 26A+. Advanced.

MCP Tool for Oracle Integration. PART 2.

REST-OIC tool versus MCP tool (using the same integration).

OCI setup+OIC setup+AI Agent Setup.

Introduction/Goal.

This document will show how to use MCP in OIC to be accessible from AI Agent studio and OIC REST in the same demo.

Part 1 (here in previous post) of this document is provided as preparation for this exercise.

Note.

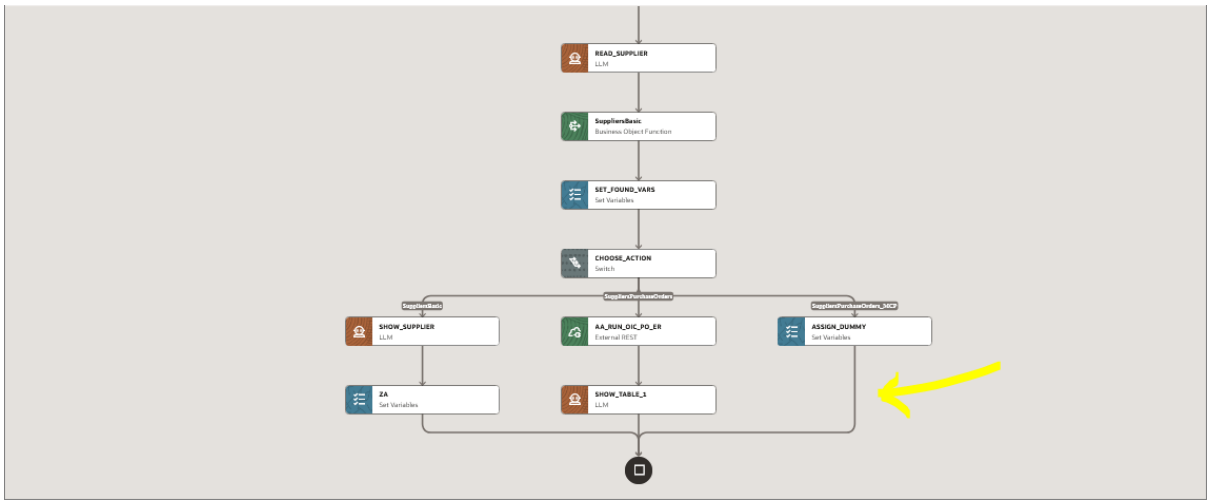
You will need access to OCI (Oracle Cloud) and OIC instances to follow this demonstration.

Part 1 explains step by step how to set up the connection and do a kind of “hello world” with a dummy integration.

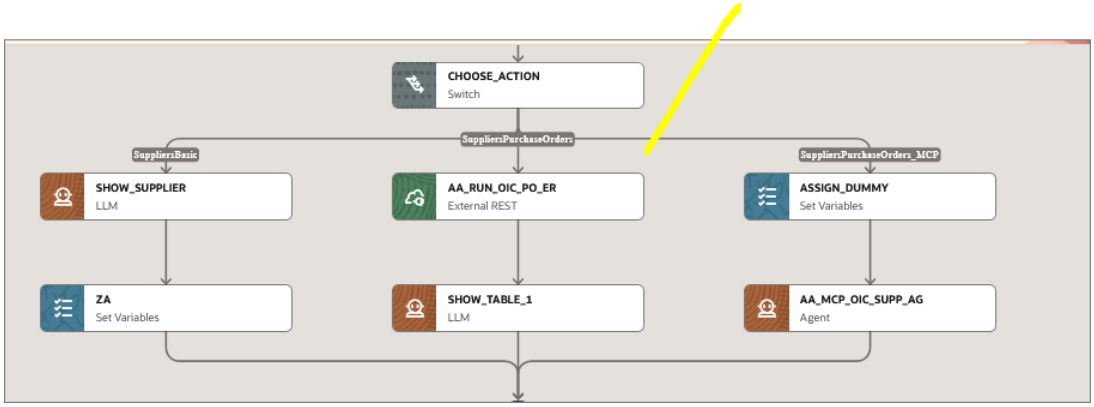
Now in Part 2, we will go a step further and will use an integration that retrieves some data depending on the parameter sent and will show this data.

Our overview of Agent team would be as follows:

Before MCP.

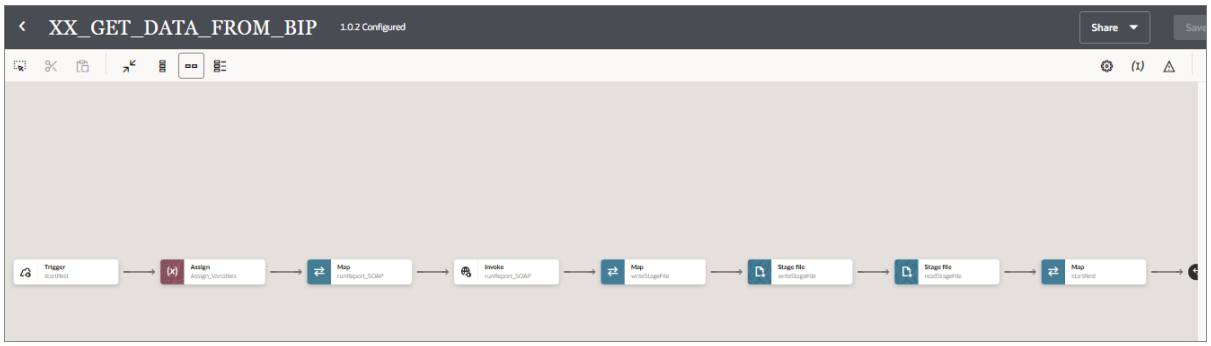


After MCP.


CHOOSE_ACTION
Switch
SuppliersBasic
SuppliersPurchaseOrders
SuppliersPurchaseOrders_MCP
SHOW_SUPPLIER
LLM
AA_RUN_OIC_PO_ER
External REST
ASSIGN_DUMMY
Set Variables
ZA
Set Variables
SHOW_TABLE_1
LLM
AA_MCP_OIC_SUPP_AG
Agent

Oracle Integration.

Integration overview.



Summary.

This integration receives a “supplier_id” as parameter, runs a BI-PUBLISHER report for some Procurement tables and returns a JSON response.

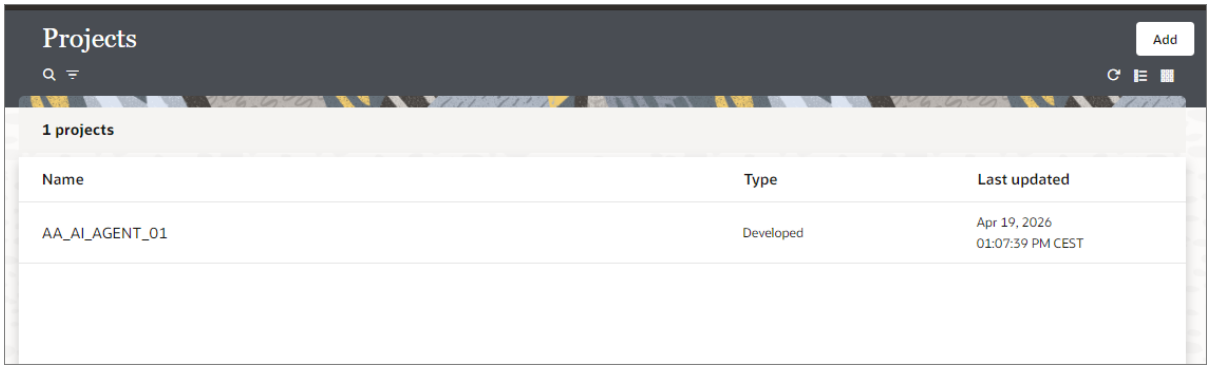
It is outside the scope of the exercise the detail to get this integration working.

Code is provided in GitHub, see below.

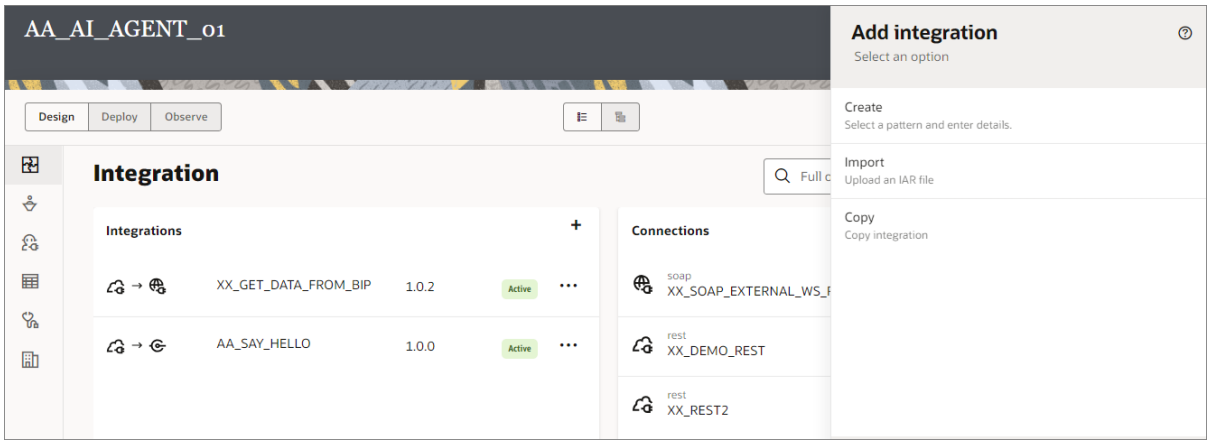
Oracle Integration. Project setup.

After we have got the integration working, we have followed these steps.

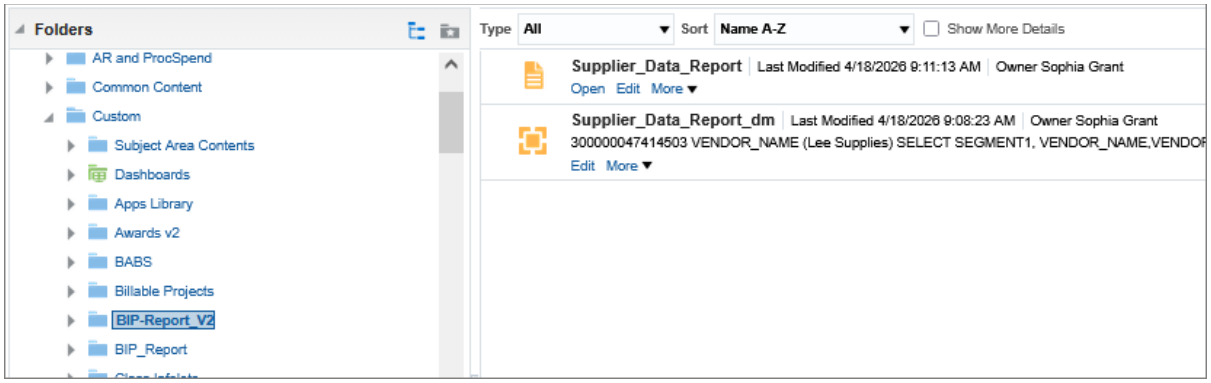
1. Open project from Part 1 in Oracle Integration.



2. Import integration.

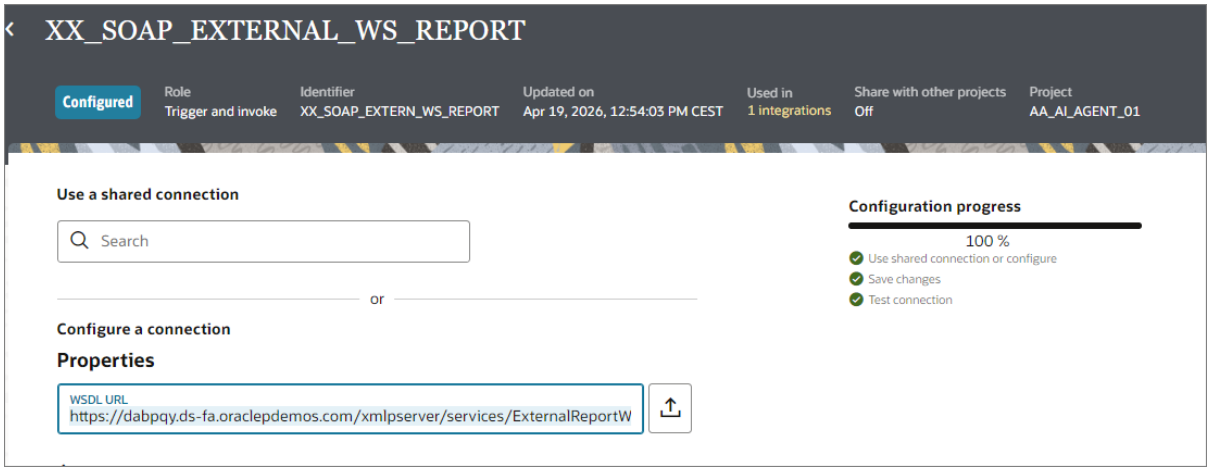


Note: BI PUBLISHER report (provided) should be in this folder in SaaS:



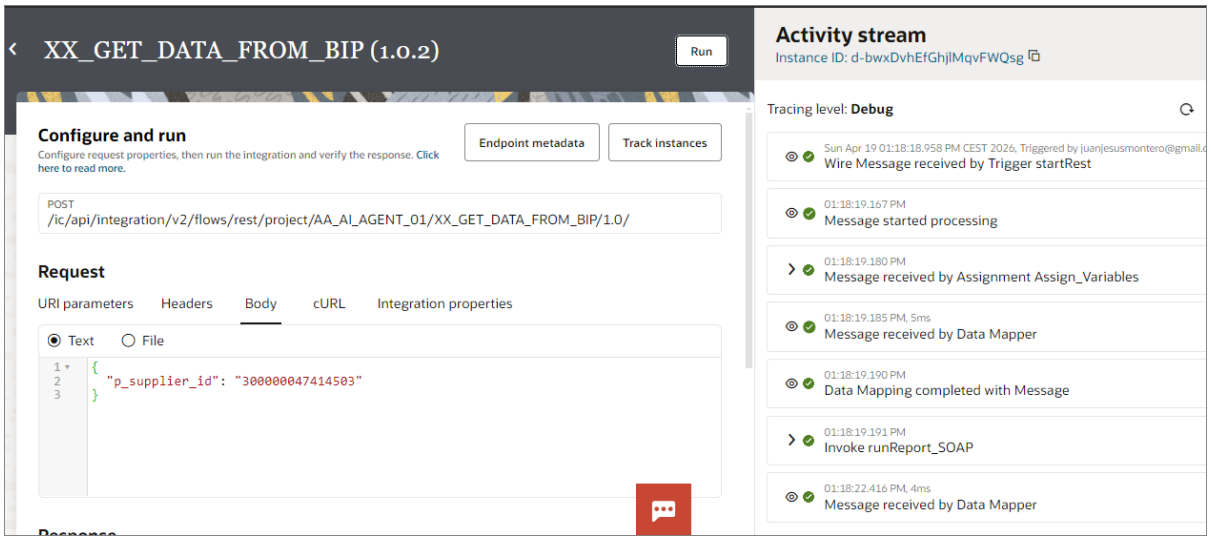
3. Adapt OIC connection.

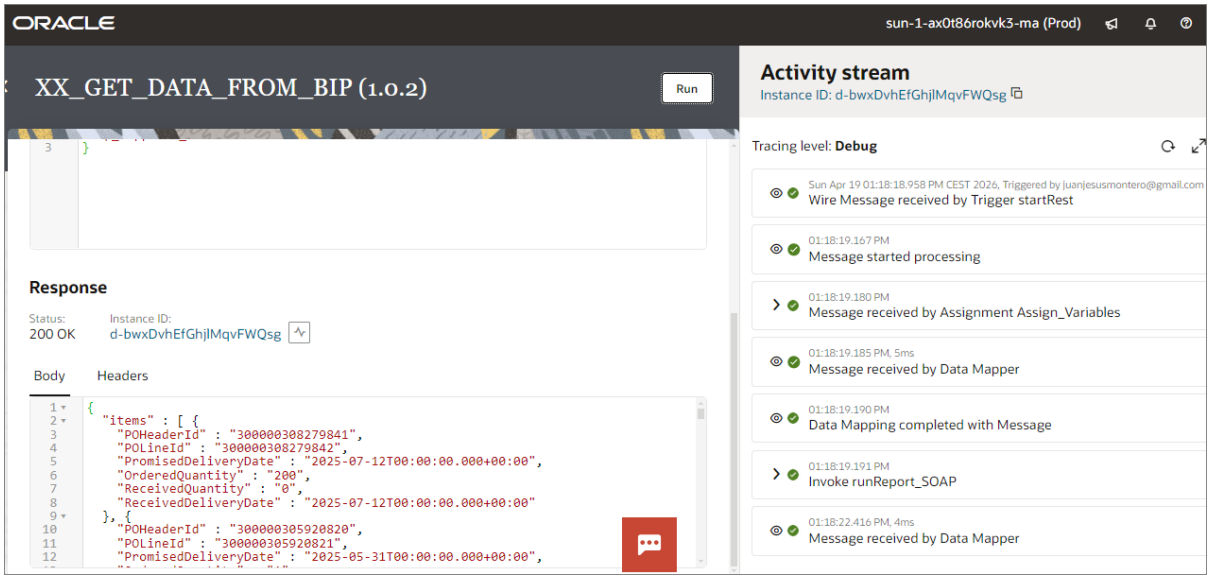
https://dabpqy.ds-fa.oraclepdemos.com/xmlpserver/services/ExternalReportWSSService?WSDL



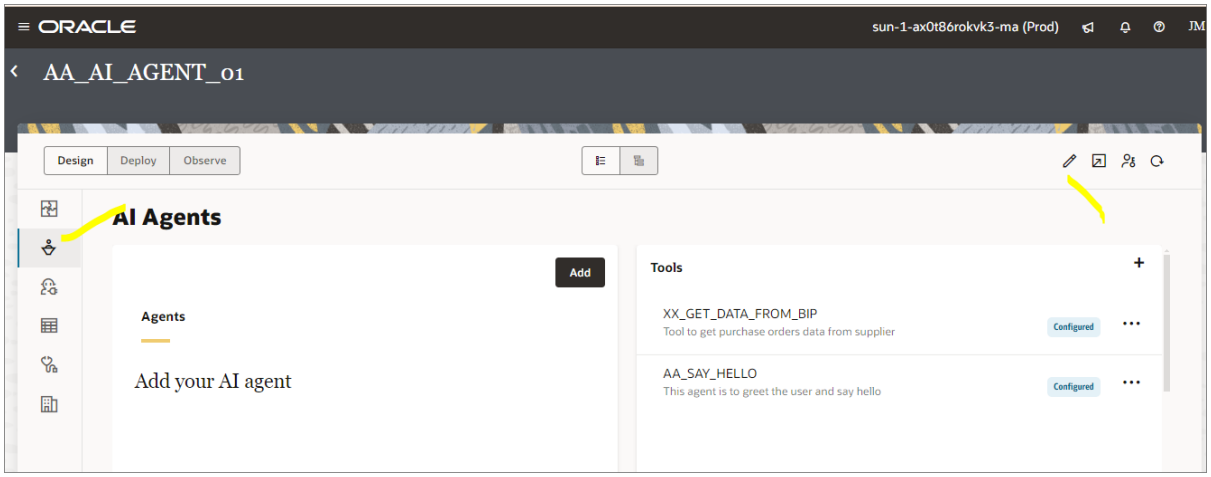
4. Activate and Test integration (provided).

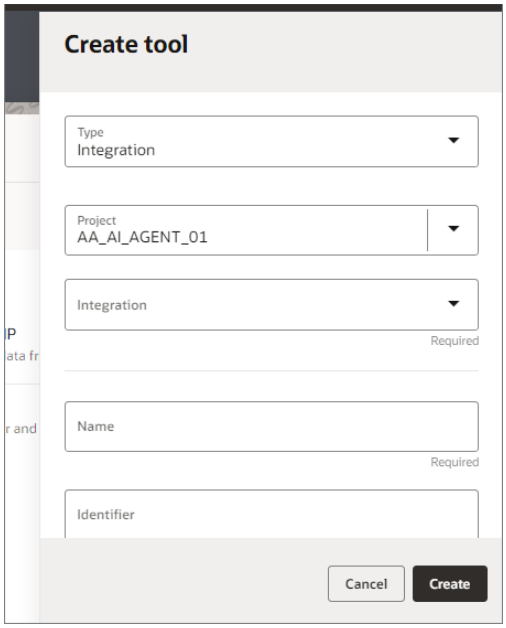
Our supplier id for testing: 300000047414503





5. Create a tool in OIC as shown, needed to access from AI Agent.

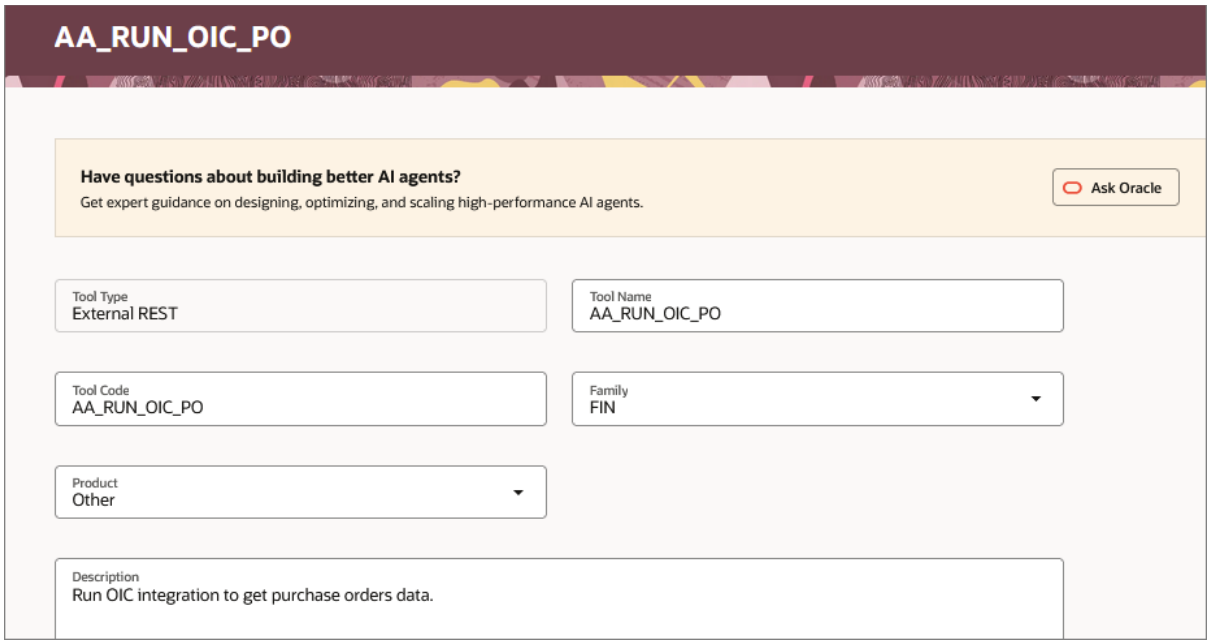

ORACLE
sun-1-ax0t86rokvk3-ma (Prod)
AA_AI_AGENT_01
Design
Deploy
Observe
AI Agents
Add
Agents
Add your AI agent
Tools
XX_GET_DATA_FROM_BIP
Tool to get purchase orders data from supplier
Configured
AA_SAY_HELLO
This agent is to greet the user and say hello
Configured


Create tool
Type
Integration
Project
AA_AI_AGENT_01
Integration
Required
Name
Required
Identifier
Cancel
Create

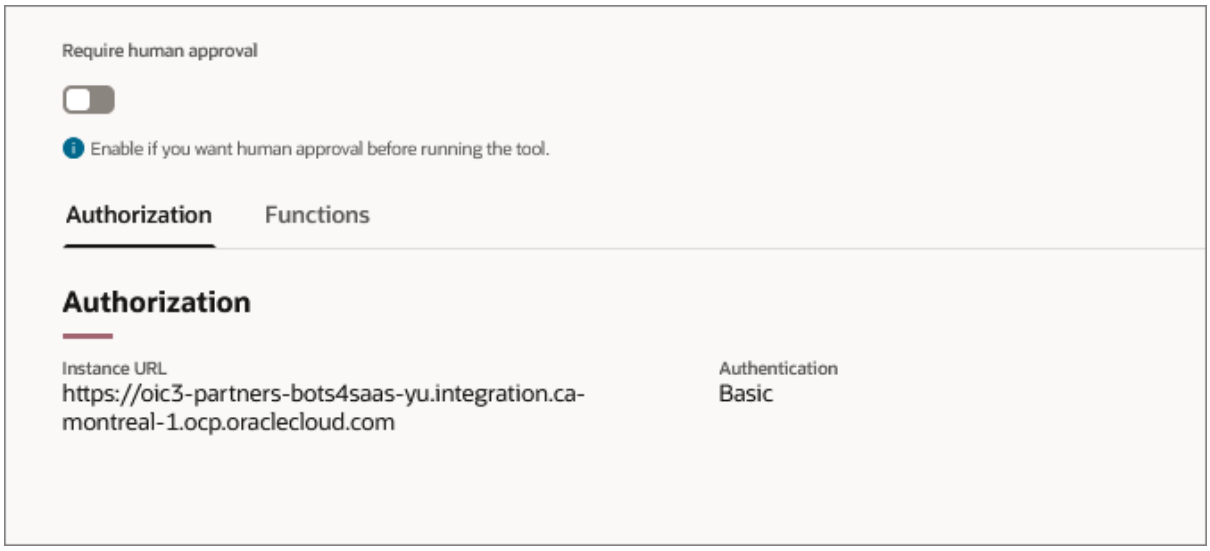
AI Agent Studio. Steps before MCP tool.

The section shows the setup needed to create a tool for REST api for OIC integration in AI Agent studio.

These are the screenshots to show the steps followed:



Authorization.



Function.

Edit Endpoint

Name
RUN_OIC
Only numbers, alphabets, _ and - are allowed

Operation Type
HTTP POST

Resource Path
/ic/api/integration/v2/flows/rest/project/AA_AI_AGENT_01/XX_GET_DATA_FROM_BIP/1.0/

Description
Run OIC integration to get purchase orders data

Body Template
```
{
  "p_supplier_id": "{p_supplier_id}"
}
```

Parameter.

Parameters Sample Queries Headers

+ Add

Name
p_supplier_id

Data Type
String

Description
Supplier id

Sample query.

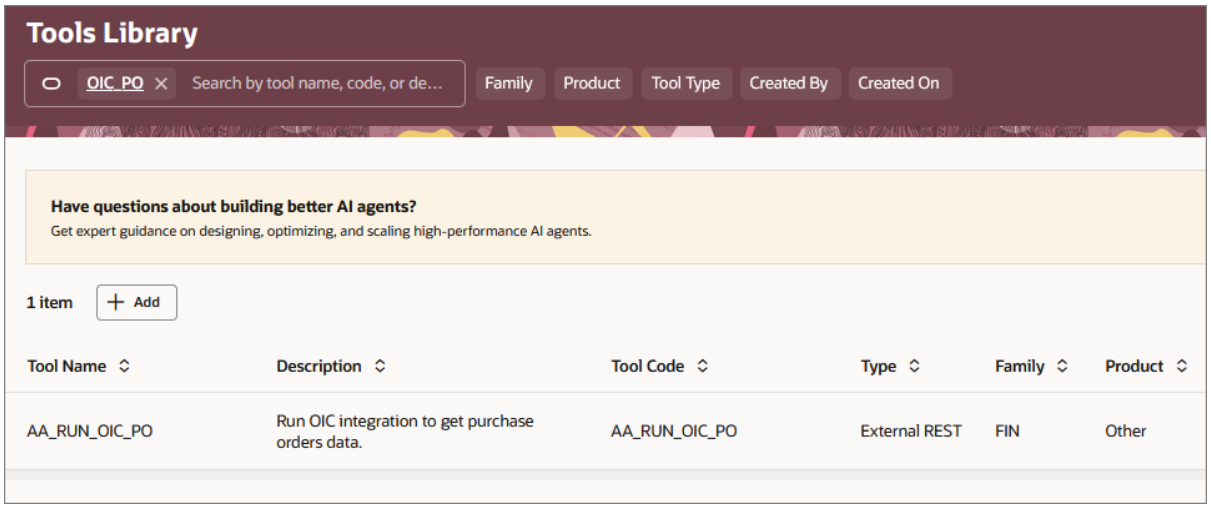
Parameters **Sample Queries** Headers

+ Add

Query
```
{
  "p_supplier_id": "300000047414503"
}
```

Description
Payload

Result.

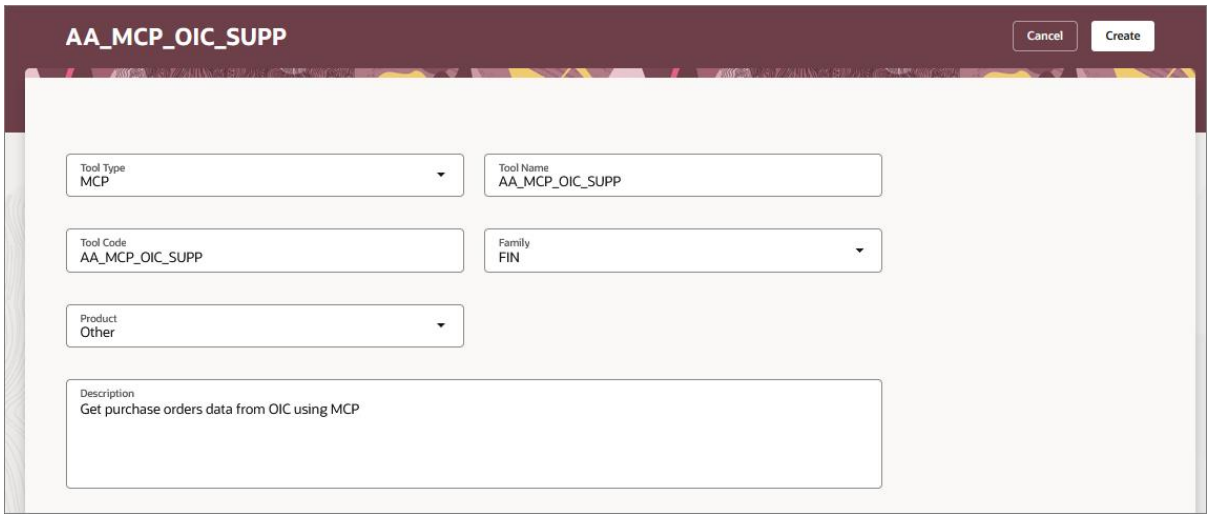

Tools Library
OIC_PO
Search by tool name, code, or de...
Family
Product
Tool Type
Created By
Created On
Have questions about building better AI agents?
Get expert guidance on designing, optimizing, and scaling high-performance AI agents.
1 item
Add
Tool Name
Description
Tool Code
Type
Family
Product
AA_RUN_OIC_PO
Run OIC integration to get purchase orders data.
AA_RUN_OIC_PO
External REST
FIN
Other

AI Agent Studio. MCP Tool setup.

1. Import provided JSON.

Note: provided JSON file “AA_DEMO_MCP_OIC_SUPP(11).json”: cover steps before including MCP tool.

2. Our steps to Create MCP Tool, before including in Agent Team.



Our data from OCI (see Part 1).

Instance URL
https://sun-1-ax0t86rokvk3-ma.integration.eu-madrid-1.ocp.oraclecloud.com/mcp-server/v1/projects/AA_AI_AGENT_01/mcp

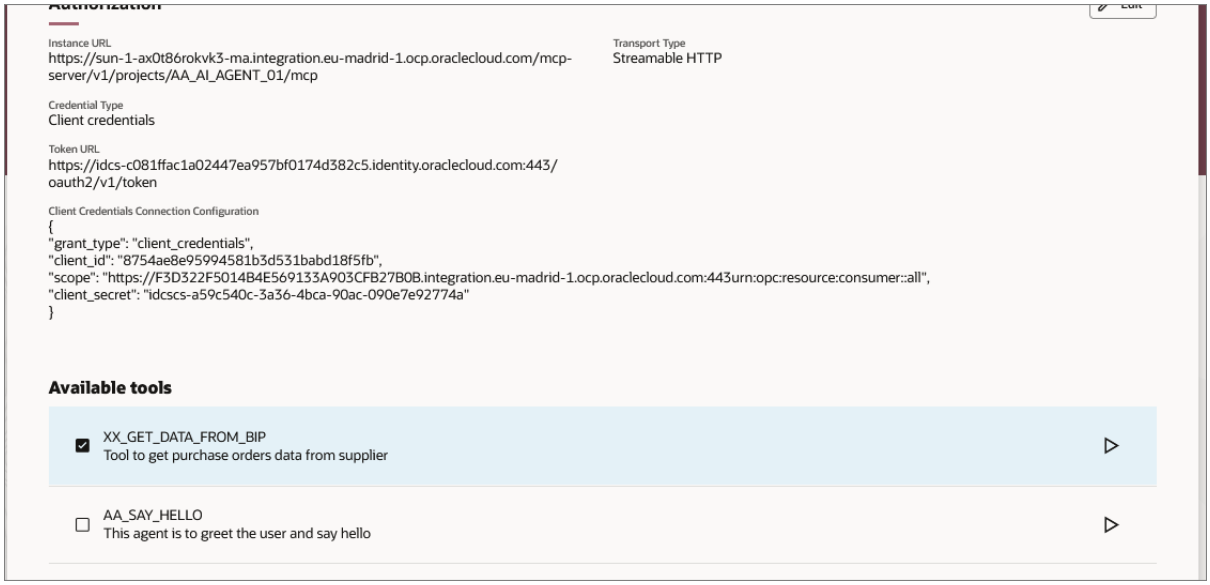
Token
https://idcs-c081ffac1a02447ea957bf0174d382c5.identity.oraclecloud.com:443/oauth2/v1/token

Client Credentials Connection Configuration.

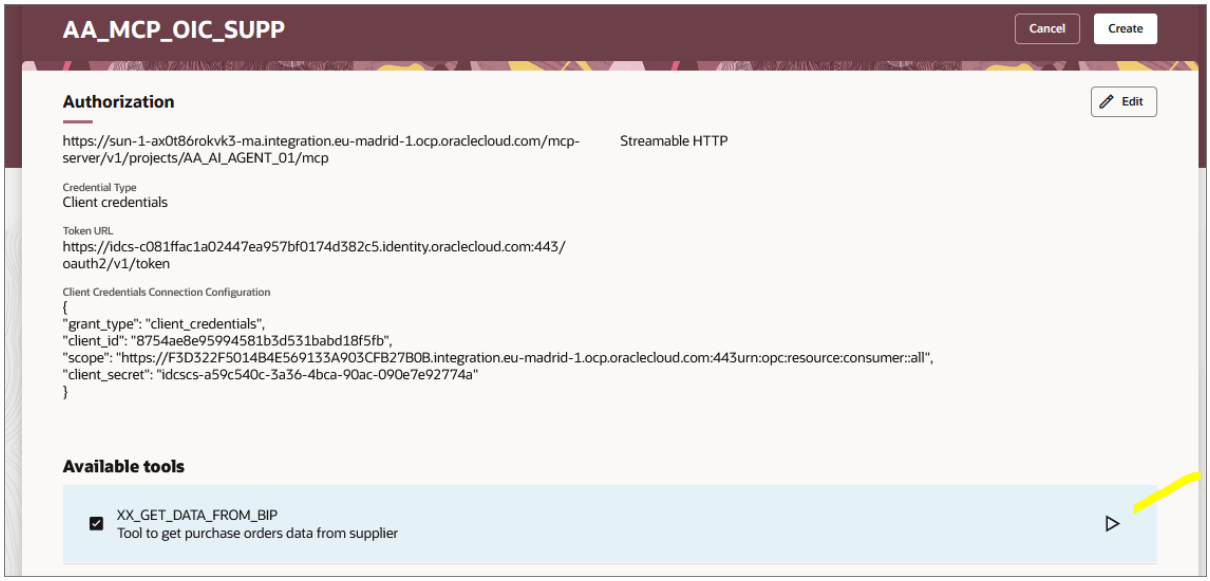
{
"grant_type": "client_credentials",
"client_id": "8754ae8e95994581b3d531babd18f5fb",
"scope": "https://F3D322F5014B4E569133A903CFB27B0B.integration.eu-madrid-1.ocp.oraclecloud.com:443urn:opc:resource:consumer::all",
"client_secret": "idcscs-a59c540c-3a36-4bca-90ac-090e7e92774a"
}

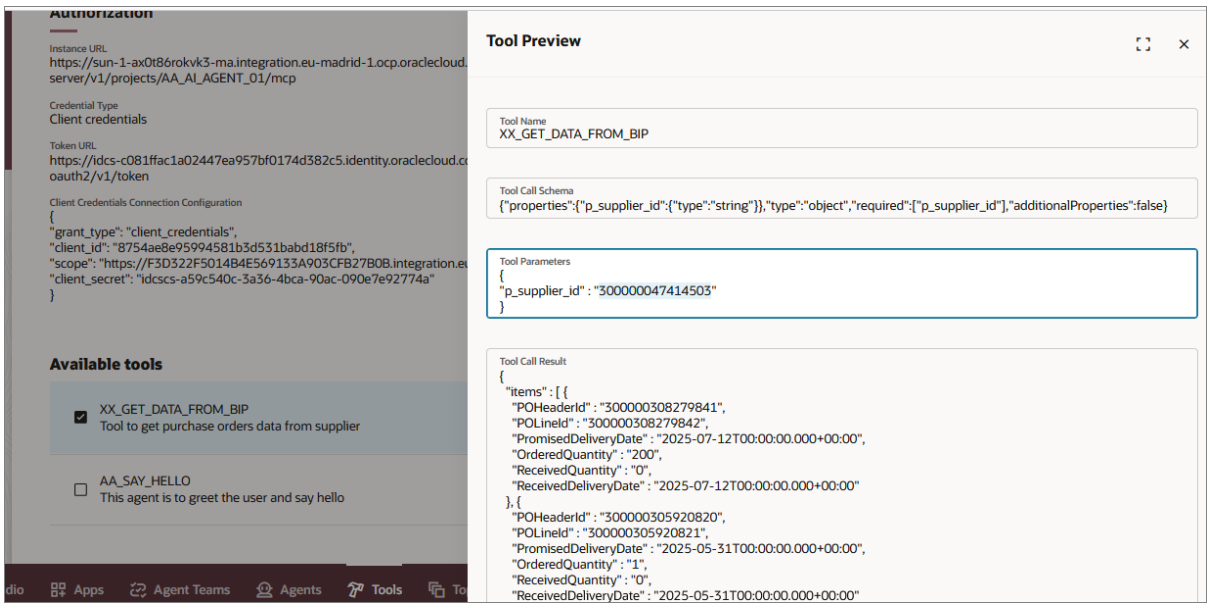
https://sun-1-ax0t86rokvk3-ma.integration.eu-madrid-1.ocp.oraclecloud.com/mcp-server/v1/projects/AA_AI_AGENT_01/mcp

Select the new integration.

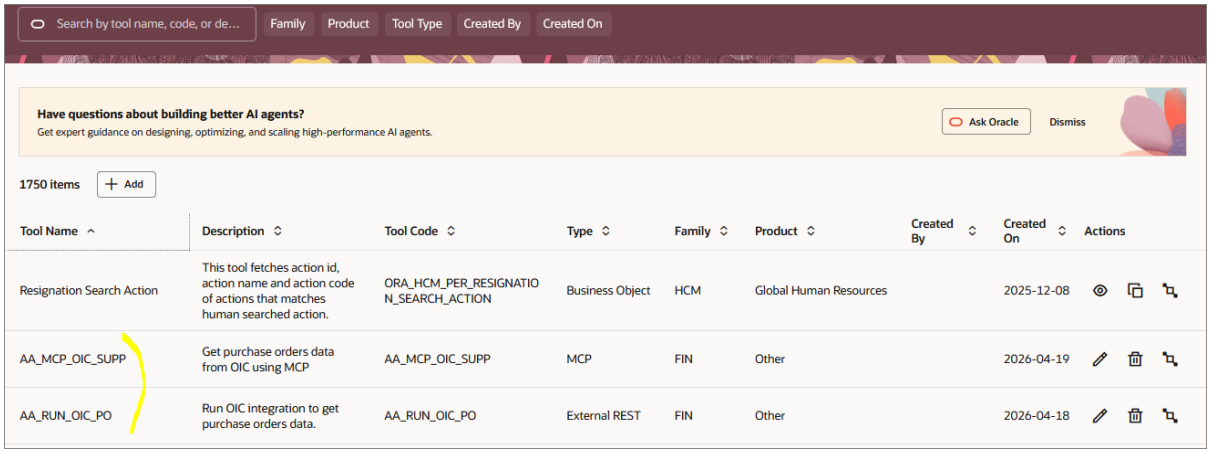

Instance URL
https://sun-1-ax0t86rokvk3-ma.integration.eu-madrid-1.ocp.oraclecloud.com/mcp-server/v1/projects/AA_AI_AGENT_01/mcp
Transport Type
Streamable HTTP
Credential Type
Client credentials
Token URL
https://idcs-c081ffac1a02447ea957bf0174d382c5.identity.oraclecloud.com:443/oauth2/v1/token
Client Credentials Connection Configuration
{
"grant_type": "client_credentials",
"client_id": "8754ae8e95994581b3d531babd18f5fb",
"scope": "https://F3D322F5014B4E569133A903CFB27B0B.integration.eu-madrid-1.ocp.oraclecloud.com:443urn:opc:resource:consumer::all",
"client_secret": "idcscs-a59c540c-3a36-4bca-90ac-090e7e92774a"
}
Available tools
XX_GET_DATA_FROM_BIP
Tool to get purchase orders data from supplier
AA_SAY_HELLO
This agent is to greet the user and say hello

3. Test.

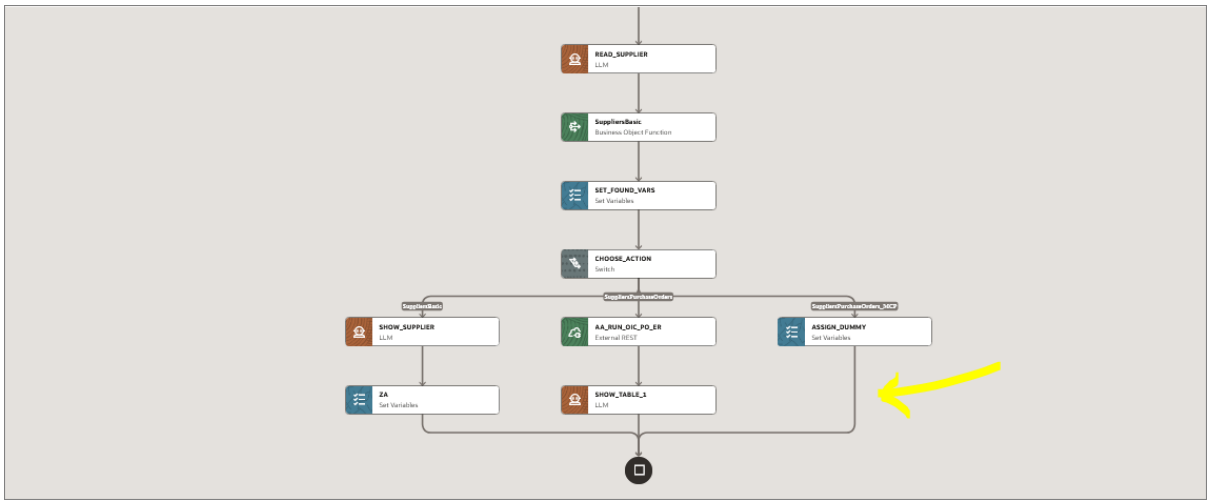

AA_MCP_OIC_SUPP
Cancel
Create
Authorization
Edit
https://sun-1-ax0t86rokvk3-ma.integration.eu-madrid-1.ocp.oraclecloud.com/mcp-server/v1/projects/AA_AI_AGENT_01/mcp
Streamable HTTP
Credential Type
Client credentials
Token URL
https://idcs-c081ffac1a02447ea957bf0174d382c5.identity.oraclecloud.com:443/oauth2/v1/token
Client Credentials Connection Configuration
{
"grant_type": "client_credentials",
"client_id": "8754ae8e95994581b3d531babd18f5fb",
"scope": "https://F3D322F5014B4E569133A903CFB27B0B.integration.eu-madrid-1.ocp.oraclecloud.com:443urn:opc:resource:consumer::all",
"client_secret": "idcscs-a59c540c-3a36-4bca-90ac-090e7e92774a"
}
Available tools
XX_GET_DATA_FROM_BIP
Tool to get purchase orders data from supplier


Instance URL
https://sun-1-ax0t86rokvk3-ma.integration.eu-madrid-1.ocp.oraclecloud.
server/v1/projects/AA_AI_AGENT_01/mcp
Credential Type
Client credentials
Token URL
https://idcs-c081ffac1a02447ea957bf0174d382c5.identity.oraclecloud.c
oauth2/v1/token
Client Credentials Connection Configuration
{
"grant_type": "client_credentials",
"client_id": "8754ae8e95994581b3d531babd18f5fb",
"scope": "https://F3D322F5014B4E569133A903CFB27B0B.integration.e
"client_secret": "idcscs-a59c540c-3a36-4bca-90ac-090e7e92774a"
}
Available tools
XX_GET_DATA_FROM_BIP
Tool to get purchase orders data from supplier
AA_SAY_HELLO
This agent is to greet the user and say hello
Apps
Agent Teams
Agents
Tools
Tool Preview
Tool Name
XX_GET_DATA_FROM_BIP
Tool Call Schema
{"properties":{"p_supplier_id":{"type":"string"}},"type":"object","required":["p_supplier_id"],"additionalProperties":false}
Tool Parameters
{
"p_supplier_id" : "300000047414503"
}
Tool Call Result
{
"items" : [{
"POHeaderId" : "300000308279841",
"POLineId" : "300000308279842",
"PromisedDeliveryDate" : "2025-07-12T00:00:00.000+00:00",
"OrderedQuantity" : "200",
"ReceivedQuantity" : "0",
"ReceivedDeliveryDate" : "2025-07-12T00:00:00.000+00:00"
}, {
"POHeaderId" : "300000305920820",
"POLineId" : "300000305920821",
"PromisedDeliveryDate" : "2025-05-31T00:00:00.000+00:00",
"OrderedQuantity" : "1",
"ReceivedQuantity" : "0",
"ReceivedDeliveryDate" : "2025-05-31T00:00:00.000+00:00"

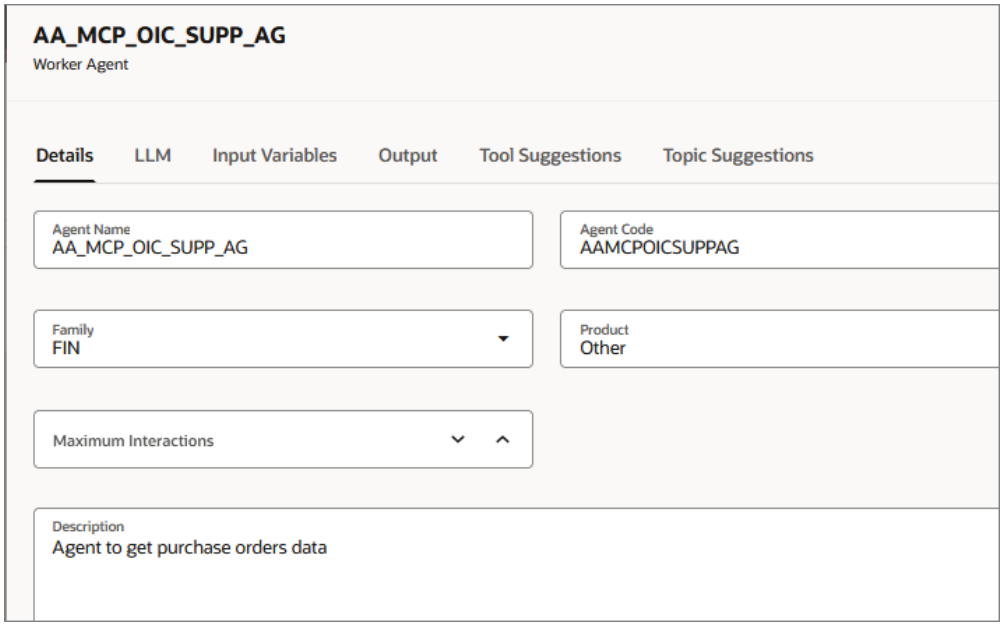
4. After click on create you will have these tools, one for normal integration and another for MCP integration.

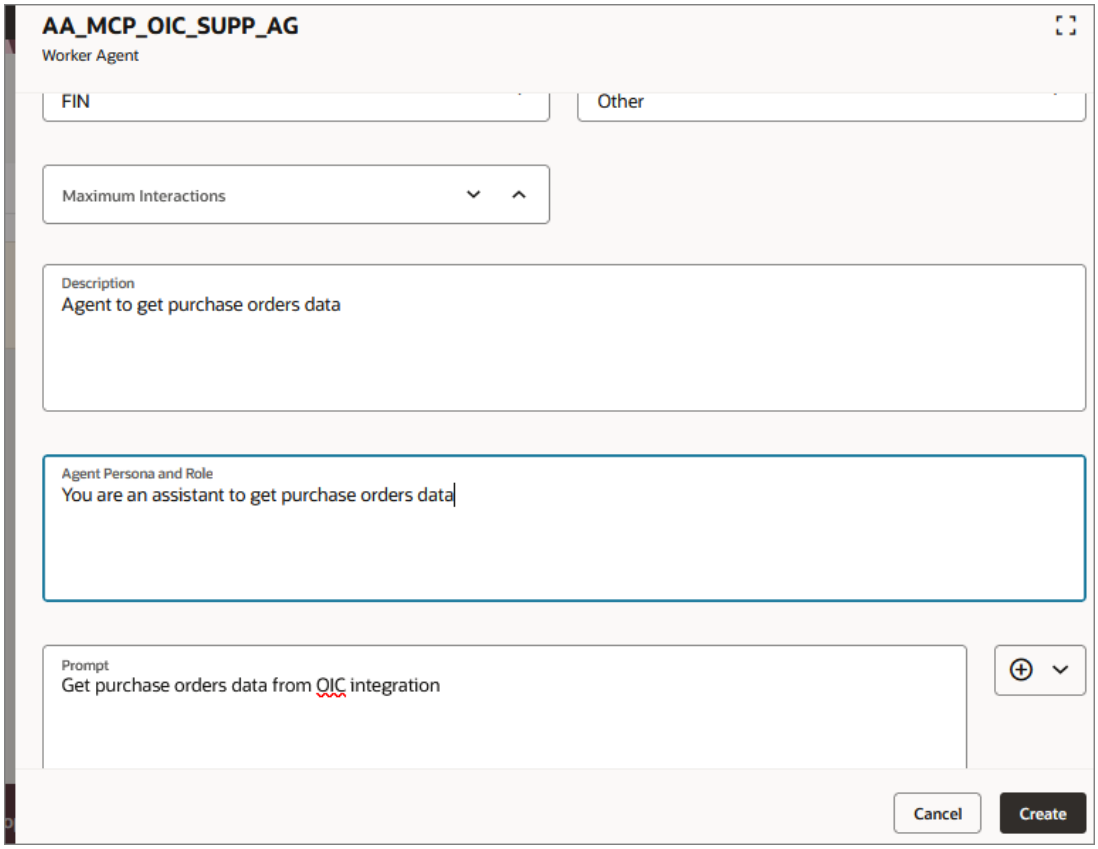


5. Now we will include our MCP tool in our Agent team.



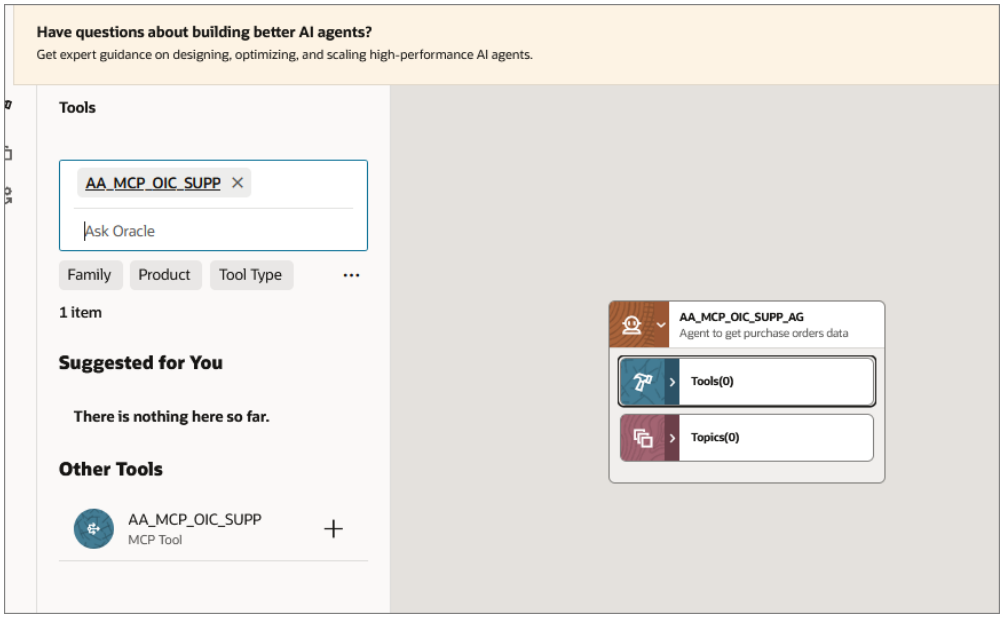
6. You will need an Agent to wrap the tool just created.





Add the MCP tool.

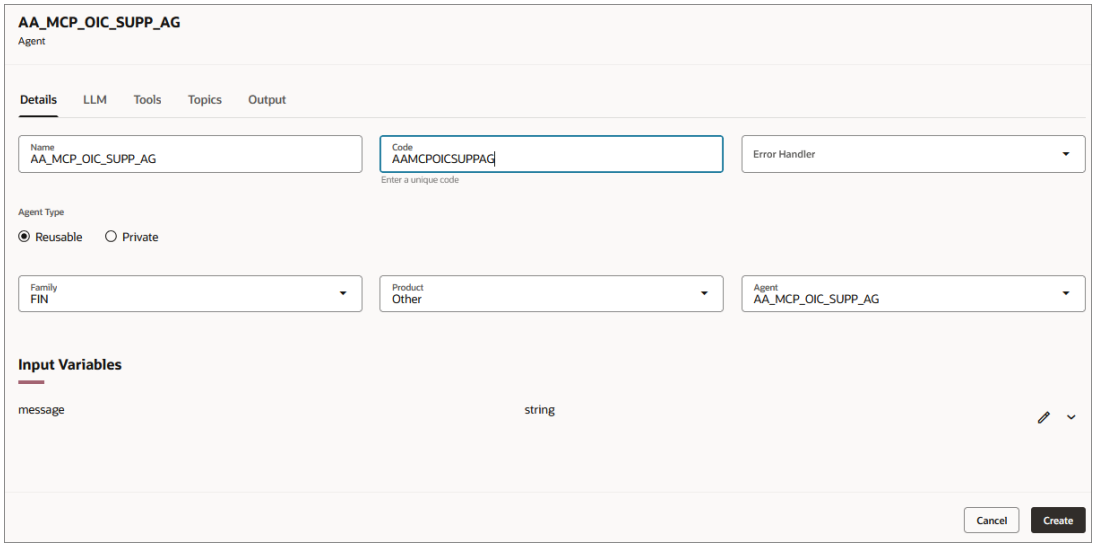
Name: AA_MCP_OIC_SUPP

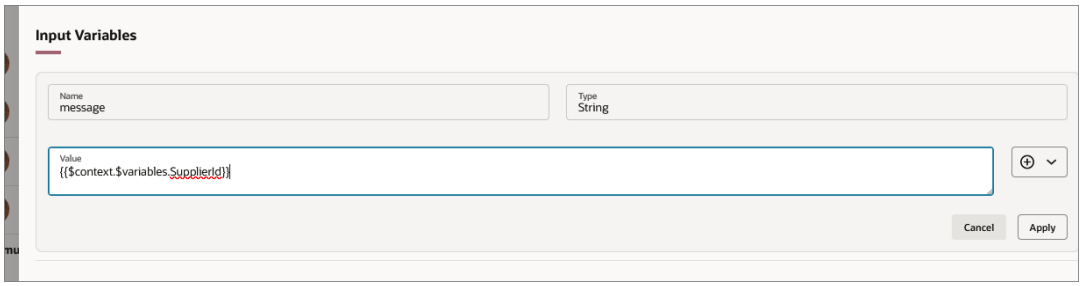


Click Create and Publish.

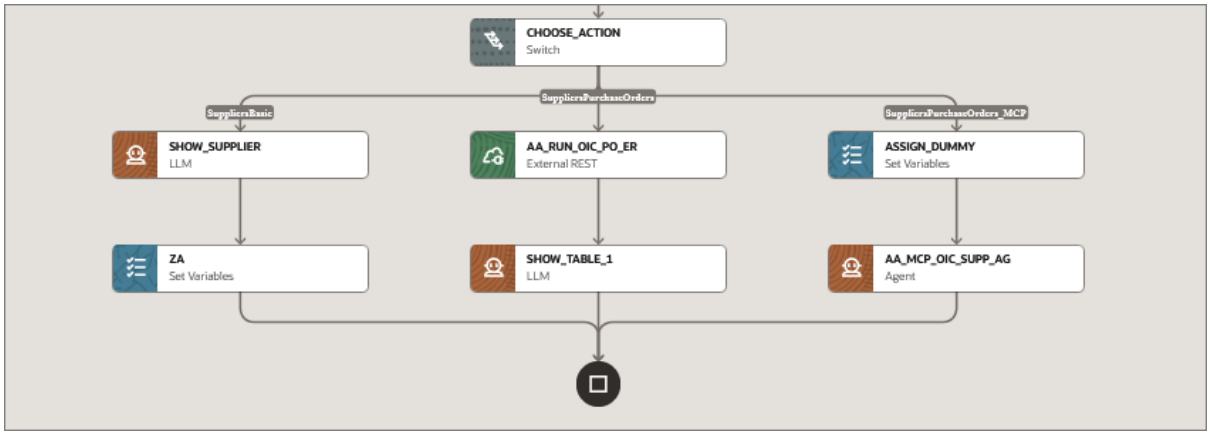
Name: AA_DEMO_MCP_AGENT

7. Now include in our Agent Team.





Now we will have this overview.



Tests.

Notice our prompt in first node LLM to guess the switch path to follow and find the supplier name from the conversation.

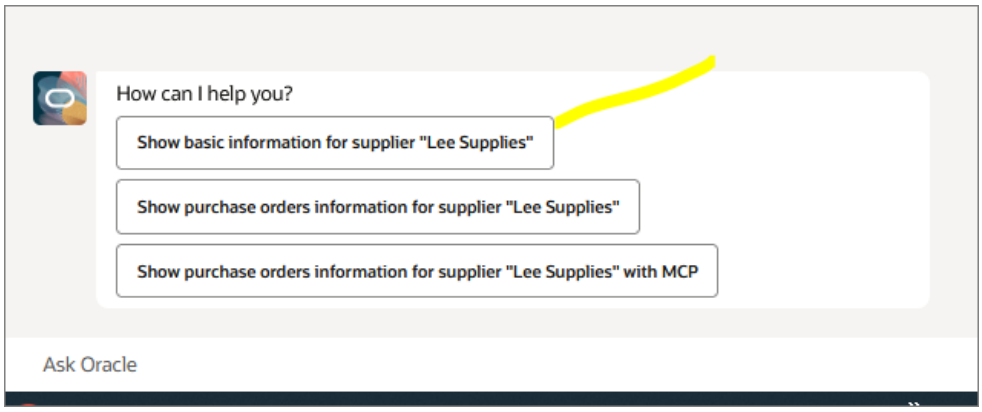
You will have two tasks.

#Task 1.
Find the supplier name from {{$context.$system.$inputMessage}} and store it in variable {{$context.$variables.SupplierName}}

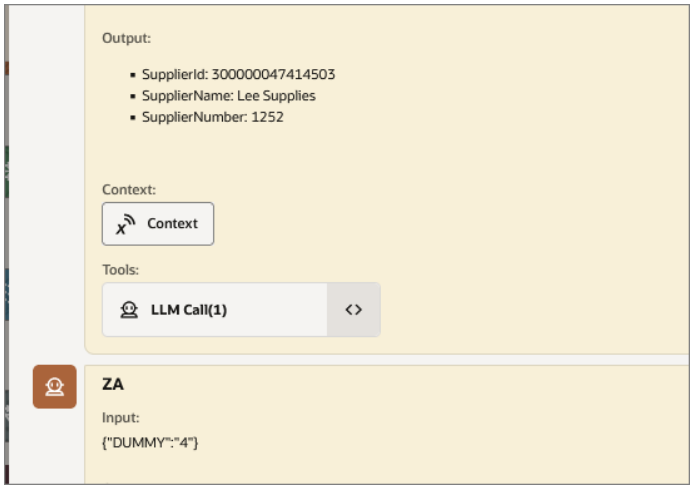
#Task 2.
Action will have these possible values:
1. SuppliersBasic.
2. SuppliersPurchaseOrders.
2. SuppliersPurchaseOrders_MCP.
You will derive the action from {{$context.$system.$inputMessage}} and store action in variable {{$context.$variables.Action}}
Select action = "SuppliersBasic" when user enter expressions like "basic", "Basic", "basic supplier info"
Select action = "SuppliersPurchaseOrders" when user enter expressions like "show purchase orders", "show purchaseorders", "show purchase orders of supplier"
Select action = "SuppliersPurchaseOrders_MCP" when user enter expressions like "show purchase orders with MCP", "show purchaseorders with MCP", "show purchase orders of supplier with MCP"

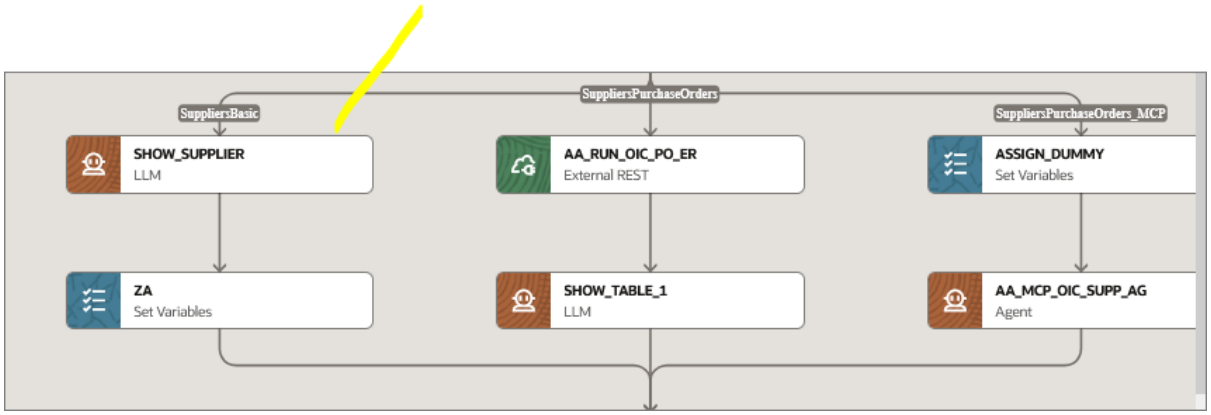
Store it in variable {{$context.$variables.Action}}

Conversation 1.

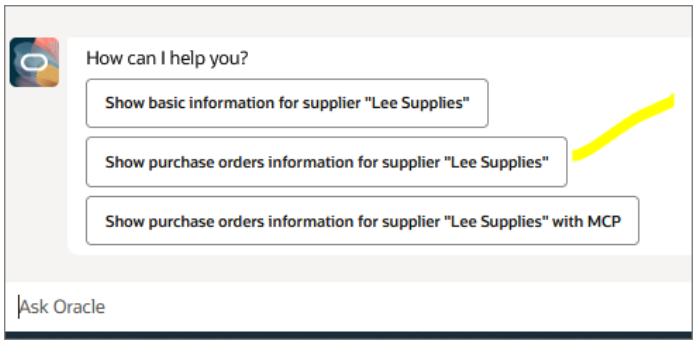


In this case, we will show basic data from supplier, using a basic Business Object.



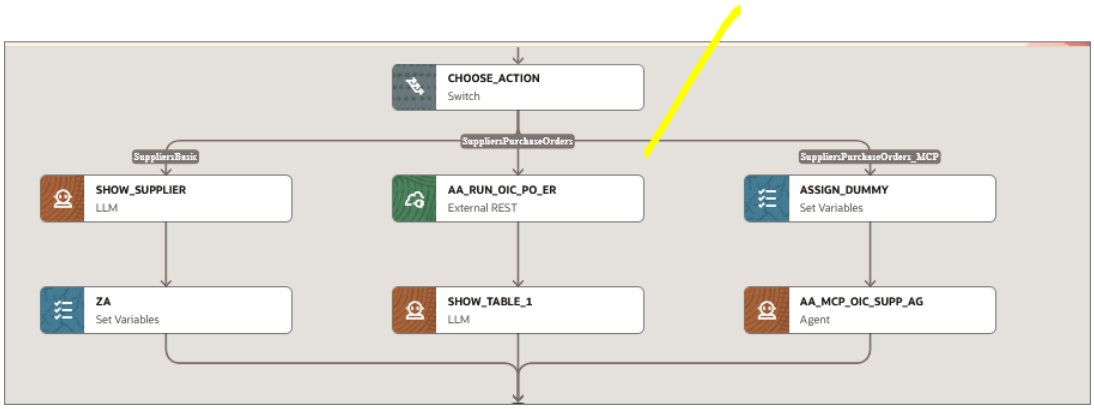


Conversation 2.

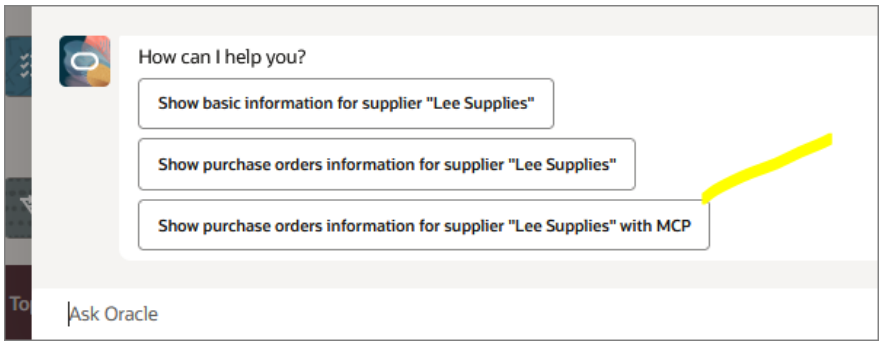


In this case, the integration called using a REST tool will be used.

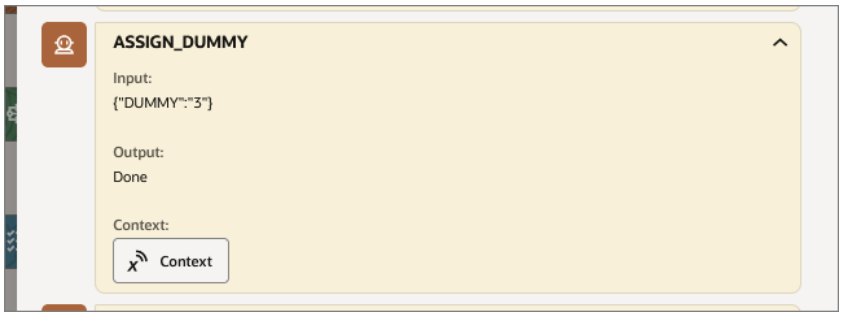
ReceivedQuantity	POHeaderId	OrderedQuantity	POLineId	ReceivedDeliveryDate	PromisedDeliveryDate
0	300000308279841	200	300000308279842	2025-Jul-12	2025-Jul-12
0	300000305920820	1	300000305920821	2025-May-31	2025-May-31
0	300000299286857	2	300000299286858	2025-Mar-02	2025-Mar-02
0	300000296213606	10	300000296213607	2025-Jan-10	2025-Jan-10
0	300000296208759	1	300000296208766	2025-Jan-15	2025-Jan-15
0	300000296208759	1	300000296208760	2025-Jan-17	2025-Jan-17
0	300000296208759	1	300000296208763	2025-Jan-17	2025-Jan-17
2	300000132880388	2	300000132880392	2016-Oct-24	2016-Oct-24
2	300000132880388	2	300000132880389	2016-Oct-24	2016-Oct-24

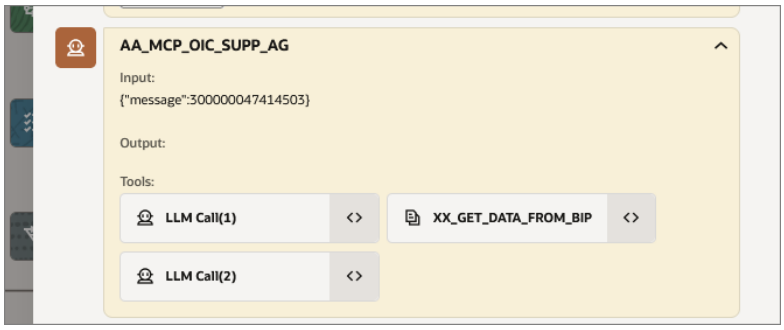


Conversation 3.



In this case, the integration called using an MCP-OIC tool will be used.





We see here we retrieve the data from MCP-Integration.

XX_GET_DATA_FROM_BIP

Latency

7.09 seconds

Input


```
{
  "tool_type": "mcp_tool",
  "tool_call_id": "call_5ZTaL76opUEXW1F0NKQA6Q2I",
  "params": {
    "p_supplier_id": 300000047414503
  }
}
```


Output


```
{
  "output": {
    "result": {
      "items": [
        {
          "POHeaderId": 300000308279841,
          "POLineId": 300000308279842,
          "PromisedDeliveryDate": "2025-07-12T00:00:00.000+00:00",
          "OrderedQuantity": 200,
          "ReceivedQuantity": 0,
          "ReceivedDeliveryDate": "2025-07-12T00:00:00.000+00:00"
        },
        {
          "POHeaderId": 300000305920820,
          "POLineId": 300000305920821,
          "PromisedDeliveryDate": "2025-05-31T00:00:00.000+00:00",
          "OrderedQuantity": 1,
          "ReceivedQuantity": 0,
          "ReceivedDeliveryDate": "2025-05-31T00:00:00.000+00:00"
        },
```


For supplier ID 300000047414503, there are several purchase orders with upcoming promised delivery dates in 2025. For example, one order promises delivery of 200 units by July 12, 2025, and another promises 1 unit by May 31, 2025. Most of these orders have not yet been received. There are also older orders from previous years, some of which have been fully received. If you need details on a specific purchase order or further assistance, feel free to ask!

show purchase orders in HTML table

Tools:

LLM Call(1)

POHeaderId	POLineId	OrderedQuantity	ReceivedQuantity	PromisedDeliveryDate	ReceivedDeliveryDate
300000308279841	300000308279842	200	0	2025-Jul-12	2025-Jul-12
300000305920820	300000305920821	1	0	2025-May-31	2025-May-31
300000299286857	300000299286858	2	0	2025-Mar-02	2025-Mar-02
300000296213606	300000296213607	10	0	2025-Jan-10	2025-Jan-10
300000296208759	300000296208766	1	0	2025-Jan-15	2025-Jan-15

Feel free to include an LLM node to get directly the HTML table.

Conclusion.

We can work with OIC integrations with MCP configuration from AI Agent Studio.

In this part 2, we have worked with an (MCP) integration that gets data from a parameter sent by the user's conversation, and also with a tool of type REST-OIC.

Feel free to build your integrations and connect to AI Agent Studio for more complex examples.

Technical

Code example	Comments
AI_Fusion_Agent 26AB MCP Tool for OIC.pdf	Document for Part 1.
AA_AI_AGENT_01_PART2.car	OIC project containing integration
AA_DEMO_MCP_OIC_SUPP(11).json	Agent team available to import in 26A+ instances. Steps before including MCP tool.
AA_DEMO_MCP_OIC_SUPP(12).json	Agent team available to import in 26A+ instances. Finished as shown in demo
XX_GET_DATA_FROM_BIP_01.00.0002.iar	Individual Integration for AI tool type REST
AA_AI_AGENT_01_PART2.car	Project for OIC
BIP-Report_V2.catalog	BI Publisher report